Section 1.7 — Absolute Value Functions — Answer Key

Math 103: College Algebra

Evaluating Absolute Values

Evaluate each expression.

1. $|2-3| + \dfrac{|5|}{|-4|} = \frac{9}{4}$
2. $-|-5| \cdot |7-1| = -30$
3. $2 \cdot -|6+3| - |-2 \cdot 6| = -30$
4. $-\frac{1}{3}|2-8| + 4 = 2$

Identifying Absolute Value Functions

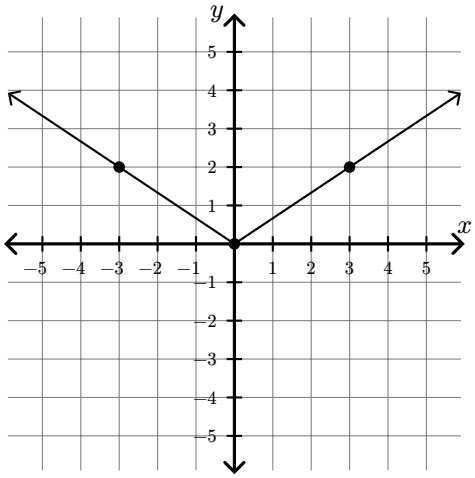
Determine whether each of the following is an absolute value function.

5. Yes
6. No
7. No
8. Yes
9. No
10. No
11. Yes
12. No

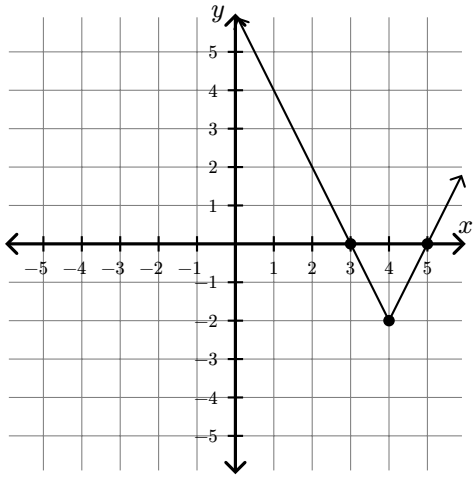
Graphing Absolute Value Functions

Graph each absolute value function. Then, looking at the graph, state the domain and range of each function.

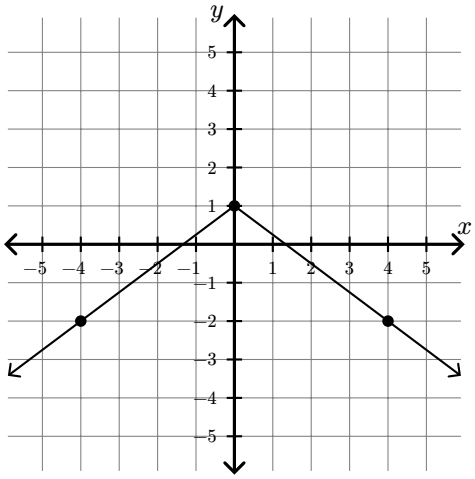
13. $f(x) = \frac{2}{3}|x|$
    $\operatorname{Dom} f = \mathbb{R} \quad \operatorname{Rng} f = [0, \infty)$



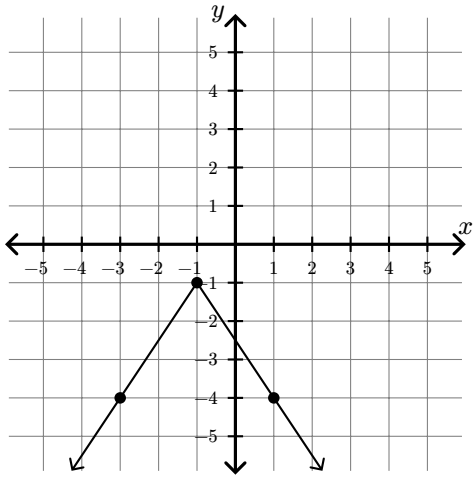
14. $g(x) = 2|x-4| - 2$
    $\operatorname{Dom} g = \mathbb{R} \quad \operatorname{Rng} g = [-2, \infty)$



15. $h(x) = -\frac{3}{4}|x| + 1$
    $\operatorname{Dom} h = \mathbb{R} \quad \operatorname{Rng} h = (-\infty, 1]$



16. $k(x) = -\frac{3}{2}|x+1| - 1$
    $\operatorname{Dom} k = \mathbb{R} \quad \operatorname{Rng} k = (-\infty, -1]$



Range and Domain of an absolute value function

Find the domain and range of each function.

17. $\operatorname{Dom} f = \mathbb{R}$
    $\operatorname{Rng} f = [0, \infty)$

18. $\operatorname{Dom} g = \mathbb{R}$
    $\operatorname{Rng} g = [4, \infty)$

19. $\operatorname{Dom} h = \mathbb{R}$
    $\operatorname{Rng} h = (-\infty, 3]$

20. $\operatorname{Dom} k = \mathbb{R}$
    $\operatorname{Rng} k = [-4, \infty)$